

Yiming Yang

Ph.D. Candidate in ECE department, Michigan Tech

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EDUCATION

Ph.D. in Electrical and Computer Engineering

Expected May 2026

Michigan Technological University, Houghton, MI, USA

- Thesis: Robust Autonomous Driving in Winter Weather: Perception, Planning, and Control Under Adverse Conditions

Master in Control Science and Engineering

Sep. 2017 - Jul. 2020

Chang'an University, Xi'an, China

- Thesis: Constrained Optimization and Distributed Model Predictive Control-Based Merging Strategies for Adjacent Connected Autonomous Vehicle Platoons

Bachelor in Automation

Sep. 2013 - Jul. 2017

Chang'an University, Xi'an, China

- Graduated from the "Excellent Engineer" honors program

RESEARCH INTERESTS

- **Perception system assessment under adverse winter conditions:** This study aims to quantify LiDAR-based 3D object detection performance degradation in snow, using a large-scale winter autonomy dataset to analyze the combined impact of snowfall intensity, range, and scene type.
- **Navigation in snow-covered & white-out conditions:** Robust winter navigation and traversability estimation by detecting the unique thermal properties of wheel tracks in visually degraded, snow-covered environments using RGB-Thermal (RGB-T) sensor fusion in white-out conditions.
- **Autonomous control on snow and ice:** Investigate vehicle performance limits on snow and ice using a neural network dynamics model and a Model Predictive Path Integral (MPPI) controller, validate on a 1:5 scale autonomous testbed to quantify performance degradation in winter.

RESEARCH EXPERIENCE

Proposal: Robust 3D Human Pose Estimation and Tracking in Adverse Winter Conditions through Thermal and VIS Fusion

Submitted Sep. 2025

- Role: Lead Author and Researcher.
- Sponsor: Sony Research Award Program.

1. SAE Autodrive Challenge II

2022 - present

- Role: Graduate Teaching Assistant (GTA) of teams: Controls and Mathworks, Safety and Testing.
- Sponsor: General Motors SAE International.
- Contributions: Architected the vehicle control interface with dSPACE MicroAutoBox III; designed and implemented the ROS-dSPACE communication pipeline; developed a safety-critical autonomy state machine with robust failure handling and safe-exit mechanisms; led HIL and VIL testing and subsystem validation on a dyno; authored key safety documentation, e.g., DVP&R.

2. The Railroad Crossing Vehicle Warning (RCVW) system

2021 - 2022

- Role: Graduate Research Assistant.
- Sponsor: Federal Railroad Administration (FRA).
- Contribution: Led the system implementation, including the configuration of VBS/RBS/RSU hardware and software, railroad crossing map creation, and field testing; developed a HIL virtual demonstration system to replicate real-world scenarios; created and presented large-scale virtual demonstrations of the RCVW application for Connected Vehicles using PTV VISSIM for the Rail Learning System (RLS) and FRA website.

- 3. Integration of Unmanned Aerial Systems Data Collection Into Day-to-Day Usage for Transportation Infrastructure – A Phase III Project** 2021 - 2022
- Role: Graduate Research Assistant.
 - Sponsor: Michigan Department of Transportation (MDOT).
 - Contributions: Executed detailed data labeling on the Williamston Road (I-96) traffic corridor dataset to train a machine-learning model for autonomous vehicle counting; subsequently analyzed the traffic data to generate OD insights and validate vehicle counts.
- 4. Microscopic Traffic Safety Analysis and macro-level transportation analysis** 2021 - 2022
- Role: Graduate Research Assistant.
 - Affiliation: Dept. of Civil, Environmental, and Geospatial Engineering, Michigan Tech.
 - Contributions: Conducted a microscopic safety analysis on the Waymo dataset by visualizing agent dynamics against road structures, extracting lane geometries to identify intersection conflict points, and generating lane polygons for precise map-matching; additionally, performed a macro-level transportation analysis using the Weijo dataset to conduct an OD study, identifying key mobility trends and peak-hour congestion zones.
- 5. From OSM to VISUM: Automated Infrastructure Generation for Transport Planning.** 2021 - 2022
- Role: Graduate Research Assistant.
 - Affiliation: Dept. of Civil, Environmental, and Geospatial Engineering, Michigan Tech.
 - Contributions: Developed an automated pipeline to convert OpenStreetMap data (nodes, ways, tags) into simulation-ready PTV VISUM networks while ensuring network connectivity, generated Points of Interest and automated zone divisions, and exported the final topology into VISUM-compatible formats (.net/.ver). This automation reduced manual network-building time from days to minutes, enabling rapid scenario testing.
- 6. From OSM to VISSIM: Automating Large-Scale Traffic Network Modeling** 2021 - 2022
- Role: Graduate Research Assistant.
 - Affiliation: Dept. of Civil, Environmental, and Geospatial Engineering, Michigan Tech.
 - Contributions: Developed an automated pipeline converting OSM data into high-fidelity PTV VISSIM models; interpreted road class, lane counts, and speed limits to model MUTCD-compliant intersections and utilized spline-based connectors for precise vehicle paths. The process generated native (.inpx) files, enabling rapid creation of large-scale, high-resolution traffic models.
- 7. Research on Construction and Site Testing Technology of Closed Test Environment for Auto-driving Electric Vehicles** 2017 - 2020
- Role: Graduate Research Assistant.
 - Sponsor: National Key Research and Development Program.
 - Contributions: Led hardware and software adaptation to transition the autonomous system from gasoline to EV architecture; developed an single-chip microcomputer based vehicle controller, and engineered and calibrated precise drive-by-wire algorithms for steering, propulsion, and braking.
- 8. QLTD Intelligent and Connected Expressway Testbase construction** 2017 - 2020
- Role: Graduate Research Assistant.
 - Sponsor: QiLu Transportation Development Group and Chang'an University.
 - Contributions: Implement V2X systems for both control - an adaptive IDM and a cellular-based remote control system; and safety - developed a V2X-based forward collision warning system. All functionalities were validated through real-world highway platoon tests using V2V (DSRC/C-V2X).
- 9. Key Technologies for Driverless Commercial Vehicle Control Systems** 2017 - 2020
- Role: Graduate Research Assistant.
 - Sponsor: Key Projects of Key Research and Development Programs in Shaanxi Province.
 - Contributions: Designed and implemented a by-wire motion-control system (throttle, brake, steering) using incremental PID algorithms; deployed trajectory tracking algorithms including Pure Pursuit and MPC;

additionally, engineered a safety-critical remote controller with override/E-stop functionality and power supply system for all onboard electronics.

PUBLICATIONS

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| 1. Tracking the Invisible: Self-Supervised Wheel Track Detection with Thermal-RGB Fusion | May 2025 |
| Yiming Yang, Jeremy P. Bos - Under review at IEEE Transactions on Intelligent Vehicles (T-IV), 2025 | |
| 2. Aggressive Autonomous Control on Snow and Ice | April 2025 |
| Yiming Yang, Jeremy P. Bos
SAE International Journal of Advances and Current Practices in Mobility (10.4271/2025-01-8040) | |
| 3. Constrained optimization and distributed model predictive control-based merging strategies for adjacent connected autonomous vehicle platoons | Nov. 2019 |
| Haigen Min, Yiming Yang, Yukun Fang, Pengpeng Sun, Xiangmo Zhao, 10.1109/ACCESS.2019.2952049 | |

TALKS & PRESENTATIONS

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|---|------------|
| 1. Aggressive Autonomous Control on Snow and Ice | April 2025 |
| Conference proceedings talk, World Congress Experience (WCX), 2025, Detroit, MI, USA | |
| 2. Deep Deterministic Policy Gradient based Cooperative Platoon Logitudinal Control Strategy | Jan. 2021 |
| Conference presentation: Transportation Research Board (TRB), 2021, Virtual | |
| 3. Research on Automatic Emergency Line Change Control Strategy Based on MPC | Jun. 2019 |
| Conference presentation, 2019, Beijing, China | |

PATENTS

- 1. Autonomous vehicle controllers** (NO: ZL 2019 2 0786357.5)
- 2. Autonomous vehicle platoon controller** (NO: ZL 2019 2 0784716.3)
- 3. Unmanned vehicle power system and unmanned vehicle** (NO: ZL 2019 2 1037928)

SELECTED HONORS & AWARDS

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| 1. Excellent graduate of Chang'an University | June 2020 |
| 2. First-class Academic Scholarship 2017-2018 | Nov. 2018 |
| 3. 2017-2018 University-level Outstanding Graduate Student | Nov. 2018 |
| 4. Best Technology Award of 2018 i-VISTA Auto Driving Challenge | Aug. 2018 |
| 5. First Prize of 2018 i-VISTA Auto Driving Challenge - Urban Traffic Scene Challenge | Aug. 2018 |
| 6. Second Prize of 2018 China Robot Competition - General Service Robot Project | Aug. 2018 |
| 7. First-class academic scholarship 2016-2017 | Nov. 2017 |
| 8. University-level first prize: Electric vehicle charging pile and operation management platform, Team award | Jan. 2017 |

SKILLS

Technical Skills: Computational: Python, C/C++ , MATLAB/Simulink for algorithm development; **Control Systems:** Real-time control design and implementation (dSPACE); **Robotics:** System integration with ROS; **Hardware:** CAN bus analysis, CAD (SOLIDWORKS), and PCB design (Altium Designer).